CHAPTER 8

THE NEXT PHASE OF EVOLUTION: MACHINE INTELLIGENCE

WELCOME TO THE AGE OF AUTOMATION AND AI

AUTOMATION IS A MAJOR PART OF OUR LIVES. By replacing human labor and intelligence with machines, we achieved a standard of living unknown even to royalty in past times. Automation and its more recent partner, cybernation, or the wedding of the computer to production, has unleashed an outflow of goods and services never before experienced. The next step, underway now, adds artificial intelligence (AI) — computer programming that attempts to simulate human attributes of decision-making and hypothesis testing along with self-correction. AI redesigns mechanical and electronic systems to simulate and improve upon human performance. As exciting as these developments are, they are only the beginning.

The way we conduct human affairs is being challenged by the introduction of computers. The Internet and the World Wide Web are providing the groundwork for the evolution of a new social direction in human interaction, bringing together vast stores of information from many different disciplines.

From the comfort of our homes, schools, offices and libraries we are now able to instantly access a world of information on the Internet and World Wide Web, interacting with people throughout the world. Current electronic mail and messaging systems reach Australia as quickly as the office next door. This extremely rapid, easy communication process changes radically how we relate to each other and how we conduct business. Information flows across the net statelessly — ignoring customs, borders, and international agreements. To those formerly wedded to the control of information, these are terrifying times.

Other developments in nanotechnology and replication offer humanity the opportunity to command its destiny to a degree never before achieved, to overcome scarcity once and for all, and to virtually eliminate poverty, unnecessary human suffering, deprivation, and perhaps even the need for work.

Where will this lead us? Will human beings eventually be replaced by the efficiency of machines? What will we do? How will we make a living?

As some fear, will machines enhanced by artificial intelligence eventually take over? Will people indeed become obsolete?

In this chapter, we probe the possibilities of the future of automation — its promise and its dangers. Through it all, bear in mind that these mechanical children can so far do only what we humans program them to do. For all their sophistication, they have none of our ambitions or failings; nor are they likely to. It is, therefore, our decision whether we use them to elevate people everywhere — or to serve our fears, prejudices, and power seeking. Therein lies our future, and the future of our technology.

For the first time in history, we have access to the information and training necessary to take charge of our own destiny. We are also fully responsible for the decisions we make and their consequences. Do we have the capacity, the will, and the intelligence to clearly think out and implement changes for our overall benefit, or will we wait for some catastrophic event to direct the future course?

For any society to improve the quality of life for its people, it must overcome the rigidity of the present. Science and technology undergo continuous modification and revision, whereas social customs, values and mores tend to remain fairly static. If the outmoded, unquestioning and emotion-driven methods used by our government and economic systems today had been applied to the sciences, we would have made very little technological progress.

The greatest fear people have regarding the coming age of cybernation is that millions will be left behind, unable to adjust to or understand the way the new culture operates. In fact, some people do fall behind or are slow to catch up during times of change. Most of us do not understand the science and technology behind the product we already use. Fewer and fewer of us work on our own cars. Not many people repair their own home computers, refrigerators or TV's. We have neither the training, the tools, nor the time to do so. But one of the interesting aspects of the new technologies is that you do not need understanding to use it. The human interface portion is so simple that former third world nations have easily made the jump from horse-drawn plows to computers — and many are now leading developers of software.

The history of invention includes all systems that enabled human beings to improve communication beyond the first, most primitive grunts. Books, radio, television, and all other forms of human communication extended our relationship to other human beings and added to our range of consciousness. The computer, like all other inventions, tends to serve as an extension of human consciousness, a brain outside of our bodies, yet indirectly connected to our nervous system, the world, and eventually the stars. The development of the electronic computer, World Wide Web and the Internet have liberated the user from many of the limitations imposed on them by governments. It is no longer a simple process for nations to shield their citizens from controversial ideas.

Although we are at the very early stages of the Internet the only existing threat to this new, unintended liberator is the attempt to control input and output of information. In

fact, some already want limitations imposed on material regarded — often subjectively — as being objectionable. Once established, any such control may be gradually extended to all areas that might threaten an existing power structure. The conditions that perpetuate these threats may not be a direct conspiracy, but the result of thousands of self-appointed guardians of the status quo.

Eventually, all social systems must extend beyond current boundaries and nationalities and achieve linkage, in order to arrive at a long-term, sustainable future for generations to come. In a relatively short time, most people will realize that a cybernated society may have a more beneficial potential for all humanity than any other invention in history. *Here we do not speak of the use of technology to advance the self-interests of transnational corporations, but to organize a global economy based on human rights and basic human needs.* This new world of both human and computer-generated directives can provide us with alternative strategies for global policies, part of a joint venture in problem-solving for the benefit of all the earth's inhabitants.

Automated machines today can conceivably perform almost any task accomplished by human beings. Since we have only two hands, machines have been designed that will far exceed the manipulative ability of any human being. As far back as 1961, U.S. industries announced that they had developed the first general-purpose automation machine at a price of around $2,500. It was called the TransfeRobot. Its swinging arm and hand were infinitely superior to any human arm or hand at a given task. It never tired and the electronic brain guiding it was not prone to wandering attention. It picked things up and put things down with an accuracy of two thousandths of an inch. In 1961 the Westclox Co. of LaSalle, Illinois used the TransfeRobot to oil clock assemblies as they sped by on a conveyor belt. It oiled eight precision bearings in a second.

Interestingly enough, this same year a U. S. Senate subcommittee on technology and automation revealed its observation that, considering the extent of automation, the total amount of goods and services required by the entire country could now be provided by ten percent of the work force that existed at that time. Ninety percent of the workforce no longer provided critical goods and products. Essentially, then, as long ago as 1961, 90% of the workforce toiled for non-essential goods and services. So-called service industries and work related to controlling and managing the use of money took the place of producing food and clothing.

The advent of cybernation could be regarded as the only real emancipation proclamation for humankind if used humanely and intelligently. Cybernation could enable people to have the highest conceivable standard of living with practically no labor. It could free people for the first time from a highly structured and outwardly imposed routine of repetitive day-by-day activity. It could permit one to actually live the Greek concept of leisure, where slaves did most of the work and citizens had time to cultivate their minds. The essential difference will lie in the fact that in the future, each of us will command more than a million slaves — but these will be mechanical and electrical slaves that will end forever the degrading use of any human being to do, against their will, the work of another. Perhaps

the greatest aid in enhancing the survival of the human race is the introduction of the electronic computer and artificial intelligence, which may very well save the human race from its own inadequacies.

As we begin building our requirements and plan for a new human society, we need commonly accepted values for clean air, water, and other elements of self-sustainment. These, along with a complete inventory of the earth's resources, will form the basis for a holistic approach to cybernated decision-making. Any change recommended by cybernated systems can also provide a great deal of information on the effects that innovative systems will have upon the entire system.

This is not a project for the distant future. Some of this work is already well under way.

The father of cybernation, Dr. Norbert Wiener, had this to say of the emerging age of non-human work: "It is a degradation to a human being to chain them to an oar and use them as a source of power, but it is almost an equal degradation to assign them to purely repetitive tasks in a factory which demands less than a millionth of their brain power." What dreams, what goals will we be able to achieve when we have the time to pursue them?

DEMISE OF THE MONETARY SYSTEM

Government and industry will continue to assign more and more responsibility for decision making to intelligent machines. Today's machines handle trillions of bits of information per second — far more than is manageable by any number of industrial or political decision-makers. They can also assemble and assign constantly updated information.

The other side of this trend is that so many people will be replaced, we will no longer have the purchasing power needed to sustain a monetary-based system that burdens the entire population and government with insurmountable debt.

As the old monetary system begins to displace more and more people by its reliance on automation, these people will cease to respect the authority of industry. The time-honored pattern of living in all industrial countries, the balancing of work and family interest, would become impossible to maintain for the majority of people displaced by automation.

When automation and cybernation are utilized to their fullest potential, not only industrial workers but also most professionals will be replaced by machines. It may surprise people when life-like, computer-generated images replace actors, entertainers, and television announcers as well. The movie Final Fantasy, released in 2001, featured an entirely computer-generated cast.

Even the most visionary speculative writers and futurists of the twentieth century would have had difficulties accepting the possibility of robots replacing surgeons, engineers, top management, airline pilots and other professionals. It is not unthinkable that machines may one day write novels or poems, compose music, and eventually replace humans in government and in the management of world affairs.

This is not a discussion about the morality and ethics of human participation, but a straightforward description of future technological trends. Nature does not subscribe to human interpretations of good or evil, or hang onto traits or species that are no longer useful. Nature operates without any concern whatsoever for the previous structures of living plants and organisms, many of which have been superseded again and again. There are no permanent structures in nature, although many human beings would like to believe otherwise, especially when it comes to their own species.

Although the technical changes of the future would be far beyond anything we could imagine today, the most profound effects would not be in the new technologies but rather in the many different ways that we conduct our lives and social institutions. As we move toward a more cybernated world, most people will no longer be needed to manage and operate this emerging civilization. Most of the world's fragmented social systems will be supported by a network of computerized centers and operations.

Today computers and AI are not capable of advising management with the most appropriate direction to take to maintain the competitive edge. Information regarding other corporate practices is not always known. In order for industries to maintain the competitive edge they cannot share their industrial processes, production techniques or business plans. Even if they did, a railway strike could stop their shipping ability. Predictability is often outside of their control. It is difficult to plan unless society controls a great many variables

Eventually, interlinked cyber-centers will coordinate the service industries, transportation systems, public health care and education with the latest data and the state of the world economy. Interdisciplinary teams of men and women comprised of systems engineers, computer programmers, systems analysts, researchers and the like could supervise, manage, and analyze the effectiveness of the flow of goods and services.

Such a world, linked together by communication networks and continuous flow lines of information and services, will provide a much higher standard of living to all people. Although today the consequence of automation and AI applied in a monetary base world economy often results in a much higher standard of living, it is only to a relatively small number of people. The advantages of newer technologies have not yet been made available to everyone.

Today most people look upon computers as being simply another, clever addition to technology, yet this facet of technology is now evolving to bring about the greatest force for social change we have yet encountered, allowing us an ever-widening range of decision-making in government, medicine, and industry. All indications are that AI will result in more significant changes for the human race than did any previous breakthrough or revolution.

As early as 1971, a single space satellite sent back to earth 400 miles of tape data that would take five competent analysts about 500 years to decipher and convert into useful information. We are rapidly approaching the time when human intelligence will be incapable of simulating the technological complexities necessary to manage a highly advanced society. Existing technologies are rapidly exceeding the human capacity to absorb and

process essential information. The human mind is far too slow and simplistic a structure to handle the forthcoming information surge. We have neither the training nor the capability to handle the trillions of bits of information per second necessary to efficiently manage the new advances.

That is why we urgently advocate a society that utilizes cybernetics not merely as a system of tabulation and measurement, but as a way to process vital information and channel it for the benefit of all humankind. Only our most capable computers can store and sort through the multitude of data necessary to arrive at equitable and sustainable decision analysis for the development and distribution of resources on a global scale.

In this cybernated global economy, mega machines directed by sophisticated AI will excavate canals, dig tunnels, and construct bridges, viaducts and dams — all based on designs that accommodate human and animal migrations and ecological dependencies — without the necessity of human involvement. Human participation will be in the form of selecting the desired ends. Human labor would no longer be required. In this society, construction techniques would be vastly different from those employed today. Self-erecting structures would prove most expedient and efficient in the construction of industrial plants, bridges, buildings and eventually the entire global infrastructure. This would not create cookie-cutter cities: the notion that large-scale, overall planning implies mass uniformity is absurd. Cities would be uniform only to the degree that they would require far less materials, save time and energy, and yet be flexible enough to allow for innovative changes, while maintaining the highest quality possible to support the local ecology — both human and environmental. Utilizing technology in this way would make it possible for a global society to achieve social advancement and worldwide reconstruction in the shortest time possible.

Eventually factories would be designed by robots for robots; the cybernated systems could also be self-programming by means of environmental feedback. Machines would be capable of self-replication and improvement of their operational range, while at the same time repairing themselves and updating their own circuitry. Since the computers and systems involved would be continuously self-monitoring, parts could be supplied and installed well in advance of any wear. The machines could operate continuously except when conducting their own maintenance and repair. In a resource-based economy, all the work of robots would be directed toward the well being of human beings. In such a society monitoring of people by machines would serve no useful purpose, except where deliberate human feedback was needed.

As artificial intelligence develops, machines will be assigned the tasks of complex decision-making in industrial, military and governmental affairs. **This would not imply a take-over by machines.** Instead it would be a gradual transfer of decision-making processes to machine intelligence as the next phase of social evolution.

This automated control could come into being when sensors that monitor the resources of the earth to support human activity are installed in every conceivable location, linked through a worldwide network of computers. Unlike the current trend toward polic-

ing human behavior, these benign monitors would serve the sole purpose of allowing us to arrive at the most appropriate decisions for both human and environmental benefit. I must state again that monitoring of personal behavior will be neither necessary nor desirable.

Artificial intelligence is already applied by industry in such areas as the monitoring of weather patterns by satellite, production control, and automation. With the further development of computerized systems, these environmental sensors and extensors can provide the necessary feedback to help us carefully determine each successive stage and develop the required analytical and decision-making tools. The degree to which such computerized systems are effective would be based upon the amount of sensors they were equipped with. We must always include the possibility of unforeseen variables in the environment such as fire, flood, hurricanes, earthquake, and other natural and man-made disasters.

An example of the tremendous potential of cybernated sensing systems may be seen in a hotel of the future. In the rare event of a fire, an audio-visual alarm instantly appears on the room's TV screen. The screen would then display a 3-D image and audio message, describing the route to take to avoid the fire. When exiting the building an illuminated line could indicate the way out.

The probable future of robotics is that the machines will undergo radical changes in their physical appearance and performance. They will behave more like living systems and be capable of making appropriate decisions within their sphere of operation. In the event of any unforeseen threat or danger to humans, they will act on our behalf. To maximize reliability and minimize the possibility of system failure, all computers can be programmed with a degree of flexibility and capable of automatic shutdown in the event of failure of any one of their parts.

It is irrational to fear machines in this benign role. Some people are concerned that there is too much emphasis on technology in this proposal. In fact, it is concern for humanity has inspired me to initiate these ideas for the redesign of a culture and to apply the best of science and technology to enhance the lives of everyone. We must keep in mind that it is not automated technology or machines we should be wary of, but rather the abuse and misuse of this technology by selfish interests. We can build rockets to explore outer space and to enhance the quality of life on Earth, or we can use them to destroy other nations. Ultimately, it is people who decide what ends these inanimate machines will serve. The aim of this social design is to apply advanced technology to produce abundance and improve the quality of life for all.

To reach a decision, intelligent people acquire information from appropriate sources and behave accordingly. Unfortunately, in the pursuit of advantage, humans acquire and route information for personal and corporate gain. Cybernated systems programmed for common concerns will do much to prevent unchecked, executive authority or abuses of power. In a resource-based, cybernated system, all decisions are based upon direct environmental, human and industrial feedback from all the cities, factories, warehouses, distribu-

tion facilities, and transportation networks. All the decisions are appropriate to the greater needs of society, rather than to those of the corporate interests.

Although many people claim to feel very uncomfortable about machines making decisions, everyone demands a weighing scale be used when purchasing any quantifiable amount of goods. To preclude a power failure in a hospital, people expect back-up emergency generators that automatically switch on with the least amount of inconvenience to the staff and patients. We are so used to machines making decisions about climate control, directing traffic lights, answering our phones, forwarding our messages, managing our calendars and the like, that we no longer consider them remarkable. Today people want and expect many aspects of modern life to be handled seamlessly and invisibly by so-called "intelligent machine decisions".

Few people think about or know how machines make decisions. Instead when additional machine use is proposed, they project their own personal agendas and emotions into the machine design. While some people proclaim their fear of machines, there has never been a single, deliberate act or plan by machines to hurt anyone. Unfortunately, the same cannot be said of human beings. Humans, not machines, use nerve gas and missiles to destroy. Even automobile accidents are mostly caused by human beings, rather than by mechanical failures — and even mechanical failures can often be traced back to human error.

It is easy to understand how readily we might accept computer decisions when we consider possibilities such as this: a man, leaving the top down in a convertible, feels a few raindrops and must pull over to raise the top, although it would be more convenient to have a system of sensors installed in the car that would raise the top automatically when rain begins. One sensor detects the raindrops, another scans the car to ensure no fingers or pets t could be injured by the mechanism automatically putting up the top and side windows. This, and much more, is technically feasible. The question that remains is: how smart do you wish your car to be?

Another example, designed and patented by the author, is a lightweight net installed at the bottom of a swimming pool. If a child falls into the pool, the net is activated to come up from the bottom of the pool saving the child if the parents or guardian are inattentive. When no one is using the pool the net remains at the surface. The net requires no human decision. The net's computer reacts to feedback from the environment.

Humans will still be relevant in this new society. After all, by and for whom are these machines designed?

Only in a cybernated world can decisions be based on the full range of relevant data available, without interference from human ego or self-interest. This could eventually provide us with the best solutions at the time to most social problems.

Today the majority of problems in computerized systems result from flawed human intervention. Computers will eventually be capable enough to design their own programs, improve and repair their own circuitry, and update information relevant to social needs.

Almost all life forms of the past — plants, animals, and even humanoid forms — have been replaced through the process of evolution. There are no permanent structures in nature. The assumption that the human being is the final product of evolution is based upon a narrow, self-centered projection. The human being is not a separate, self-sufficient entity: it is integrated into and dependent on nature to survive.

It is arrogant and unrealistic for us to believe that man is the final product of evolution. More and more, we see the merging of human ingenuity with machine intelligence. How many lives have been helped by the use of artificial limbs, joints, hearts, skin, and so forth? How much pure information, uncluttered by human frailties, is processed by computers every day? The next stage of evolution must surely be the merging of biological systems with man-made systems.

It takes many millions of years for living organisms to evolve, and additional thousands of years to shed vestigial organs. The human increased in size and complexity due to the many new convolutions in its neuronal development. The enlargement of the brain accompanied a corresponding increase in the capacity for associative memory.

The development of electronic systems required a different "evolution". Early computers were huge — room-sized — but now capability of information storage systems increases while the hardware occupies less and less space. Another major difference in technology is that non-functioning parts are eliminated or replaced immediately. In human systems old ideas may be retained far beyond their usefulness and new ideas shunned. In artificial intelligence and computerized systems information is updated rapidly, appropriately and constantly. Today it takes years to transfer acquired knowledge to another person, whereas in computerized systems information can be transferred instantaneously to an entire global network.

Most of us cling tenaciously to our old habits of thought, but technological progress has little use for custom and tradition. Human systems are subject to neural lag, and tend to revert to the familiar. Neural lag may be defined as the tendency to resist forming new and more appropriate associative patterns in favor of old, familiar ones. For example, during the early stages of automotive development, a statue of a horse's head was mounted to the front hood, and the rear portion of the car retained a coachman's seat.

In the field of electronics, if neural lag had prevailed, the industry would have failed to achieve its many technological innovations. To remain at the forefront of technology one must be capable of updating one's methods, discarding outdated technology, and examining new paradigms.

When we examine our present social design we find that it does not keep up with the rate of change required to take advantage of this accelerated pace of information and innovation. The person of today thinks in terms of having to get a job in order to support himself and his family. With the limitless possibilities of our technology today, this could be considered an example of neural lag. The adherence to a mentality of scarcity is another.

No matter how strong our fear or resentment of social cybernation, the process is already underway. In all branches of industry, medicine, agriculture, and technology computers are being assigned the role of decision-making. As we outgrow the need for human participation, whether it be in the military, the marketplace, or eventually government, more and more tasks will be assigned to artificial intelligence.

Although all politicians, decision-makers, educators, humanists and the literary community will probably resist this level of cybernation, the greatest level of resistance will come from the general public, attuned as they are to being directed by other humans for thousands of years.

And yet cybernation will prevail. As an old Chinese proverb has it: *The dogs may howl at the moon, but the moon will continue on its honorable journey.*

MACHINE EMOTIONS IN A CYBERNATED SOCIETY

Let us look at the concept of human emotions in machines. Imagine an automobile with computerized emotions linked to built-in, feedback mechanisms. This car is equipped with a pendulum under the hood that when an abrupt turn is made prompts it to respond via its speaker system, "What are you trying to do, destroy the car and everyone in it? Where did you learn to drive? Have you no regard for others?"

Worse yet, what if the car suddenly decided it didn't like you in the middle of that abrupt turn?

Of course, this is preposterous. And yet, how often does such an approach fail when we use it in an attempt to modify the behavior of others?

What purpose, indeed, would emotions serve if incorporated into the design of intelligent machines? Machines have no emotions. They do not feel ambition, love or hate. They do not seek power over people or harbor any repressed desire to harm or enslave anyone. They will never make demands on their users or seek revenge if misused. They will not hold a grudge, complain or manifest guile or deceit. These are flawed human traits.

With no understanding of or sensitivity to human emotions such as love and trust, machines guide aircraft, ships and spacecraft to their destinations and make the appropriate decisions to avoid troublesome weather conditions. With no concept of charity, the machines provide an abundance of food and preserve it by refrigeration. They heat and cool our homes. They sound alarms in the event of fire, warn us when hurricanes and tornadoes threaten. They order parts for machines before they wear out. Although they do not hold the hand of a distressed person, they warn us about toxic gasses in the environment.

Perhaps human emotions would be the only attribute, if given a choice, which machine technology would reject outright. When one thinks about it, the fact that machines have no emotions may in some ways make them superior to human systems — certainly so when the task requires immediate response and dispassionate weighing of options.

We could put pressure sensors in automobile tires so that they would be able to maintain their own required pressure with a built-in pump. We could program them to slow

down automatically to 15 mph when their monitors detect a school zone. If there is a child or pet behind the car when backing up, the car would automatically stop.

It makes more sense to design built-in standards of performance that operate the car, rather than indulge in the useless exercise of trying to alter the behavior of the driver with verbal abuse or stern admonitions. This same system could be applied to all aspects of the electrical, mechanical, and computerized cybernated world of the future, including human communication.

It is not a matter of the machine "caring" about the results of its actions. It is a matter of designing the function of safety into the machine. It is not emotions that are needed by machines — it is responsibility to the humans they serve. What we require of them is the ability to act intelligently with respect to human welfare and make the appropriate responses to a wide range of situations. If all of this could be accomplished without emotions, does this not raise some interesting questions about a number of human emotions?

NANOTECHNOLOGY TO COME

Nanotechnology will eventually control and direct the building of molecular structures, atom by atom, into any molecular configuration we desire. By means of such a process we will be able to rearrange matter and eliminate all shortages forever.

With the application of sophisticated technologies such as atomic and molecular replication, we might be able to reengineer the natural processes with advanced robotic manipulators that utilize phase array teletactile communication. 'Phase array' refers to the control and manipulation of light to generate three-dimensional images that appear solid. 'Teletactile' refers to the ability to impart the sensation of solidity and touch to a merely *transmitted* object. This advanced form of tele-communication will create a virtual simulation one can see, feel, hear, smell and touch.

Although such instantaneous technologies may be difficult for most to comprehend today, it is no more than an extension of current technologies, and similar to the way that color television images can be transmitted to any part of the world today. The major difference is that the image and sound will be transmitted three-dimensionally and feel solid to the touch.

The step beyond involves directly replicating an object, rather than merely simulating it. On earth, this might be accomplished instantaneously and could eventually eliminate the necessity of our cumbersome system of transporting objects from place to place by land, air, or water. Beyond the earth, it could be the future transport system from one planet or galaxy to another. Such space transportation, though achieving speeds inconceivable by present standards, would likely not be instantaneous, the rate of travel depending on the distance of the transmission.

As nanotechnology advances, machines could have a transmorphic capability, able to change their shape to the most efficient form to accomplish any given task. Such machines

could constantly assess conditions as they perform their tasks and "morph" into a more appropriate configuration as necessary.

To help understand this process, imagine evolution as a series of rapid successions, as opposed to the millions of years it takes to accomplish organic changes. Machines as described above could instantaneously rearrange their molecular structures to best serve human needs.

Of course, such machines will not look like conventional machines, any more than a microchip looks like a phonograph record. They would be as different from our present concepts of machines as humans differ from primordial life forms. Today, living systems conform to the world by natural process or they perish. In the future, machine systems will adjust the world according to the specifications set by an emerging culture, a culture that will, hopefully, be dedicated to universal human and environmental well-being.

THE DOWNSIZING OF THE GODS

AI will eventually supplant antiquated notions about gods and demons. As our own powers increase, there will surely be a corresponding decrease in humankind's tendency to seek answers and solace in religion or superstition. For example, Nature or the gods take thousands or millions of years to accomplish the slow process of evolutionary change. Modern technology can, in an instant, reproduce all the information of recorded history. With advances in nanotechnology we may some day be able to instantaneously arrange matter into any desired configuration.

Reengineering of genetic code may enable us to reduce or eliminate genetic disease and defects, perhaps even reproduce organs, bones, or tissue that is less likely to fail or be subject to disease.

Our relationship to the space-time continuum could also be modified. For example, we may be able to travel through time and space without the need for interstellar transport systems, to project human intelligence almost anywhere in the universe — in other words, unknowingly we are evolving toward a reinvention of our gods. We may even, in the future, find that we have outgrown them.

To those who might feel threatened by such concepts, it is not intelligence that we must fear, but ignorance.

IN CONCLUSION

We have arrived at a time when mathematical logic and computers can assist us in unraveling many of the processes of human thought, our growing understanding of which will enable us to enhance future computer technology. Our existing economic and political structures and processes no longer provide the support needed to keep up with and implement changes in technology. Their focus on profit, secrecy, and competition runs counter

to the possibilities for positive change afforded by the current, democratic broadening of technology, for instance the Internet with its spirit of collaboration and open exchange of information. It is time to explore new social and economic structures.

In time, the computer and social cybernation may be thought of as the only means of social management that are entirely free of selfish motivation. This may be the most humane approach to our human dilemmas.

We require a global perspective, international cooperation and planetary planning in terms of available resources. This planning must directly relate to the carrying capacity of the earth to meet the needs of all people. This can best be accomplished by using a constantly updated, computerized model of our planetary resources in a resource-based economy.

Advanced social systems do not require scientists or governments to tell them how to operate. The ultimate decision-making authority represents the expression of all humankind. This applied vision of science can serve the common good, a goal that has eluded human civilization for centuries. Although this revolution is still largely in the future, the possibility of a better life for all the inhabitants of our planet will depend, ultimately, on the choices we make today.

ROBOTS CONSTRUCTING ROBOTS

These "multi access industrial robots" utilize vast information resources, which enable them to receive commands via satellite up-link or on-site. They can also be designed to take appropriate actions in the absence of human directives by combining an array of Micro Electro Mechanical Systems (MEMS) and sensors and receivers with sophisticated decision-making circuits and artificial intelligence programs. They will be capable of handling a wide range of industrial production tasks, and will even be able to upgrade their level of service and replace their own parts. When necessary, these mega robots will communicate with one another and coordinate the logistics and delivery of the required materials for each project.

NANOTECHNOLOGY

The future of nanotechnology offers enormous potential. Nanotechnology combines optics and lasers, and will eventually enable us to assemble matter, atom by atom, into whatever molecular structure is needed. Nanotechnology will lead to a sub-microscopic revolution in all fields, including the way in which we conduct human affairs.







MEGA-EXCAVATION MACHINES

This scene depicts a laser excavator of the future. Such devices, directed via satellite, would be capable of fusing the earth beneath it into a molten, magma-like material, thus contouring the earth to aid in the construction of canals, roads, and waterways.



AUTOMATED TUNNEL ASSEMBLING MACHINE

Tunnel segments float down canals by the use of very large flotation devices. This automated tunnel-assembling machine lifts the prefabricated segments and places them in their required positions. Once completed, the tunnels are used for high-speed, mag-lev transportation.

THE CONSTRUCTION OF TOWERS

These towers are designed specifically for regions where earthquakes are prevalent. Such cable-suspended structures easily withstand a wide range of movement, stresses, and strains. Circular towers radially arranged in a city design may be self-erected efficiently and rapidly around a central core that houses elevators and all other household utilities. The translucent windows serve as photovoltaic generators, and the intensity of the light entering the windows can be varied electronically. All window cleaning and maintenance is automated.





MASSIVE LIFTING CRANE

This multi-function crane is designed to lift freeform structures and position them on foundations and stilts or transfer them to vertical lift systems that place the units on towers. Upon completion of their task, these self-erecting cranes can be automatically disassembled into a compact form for easy transportation to the next assignment.



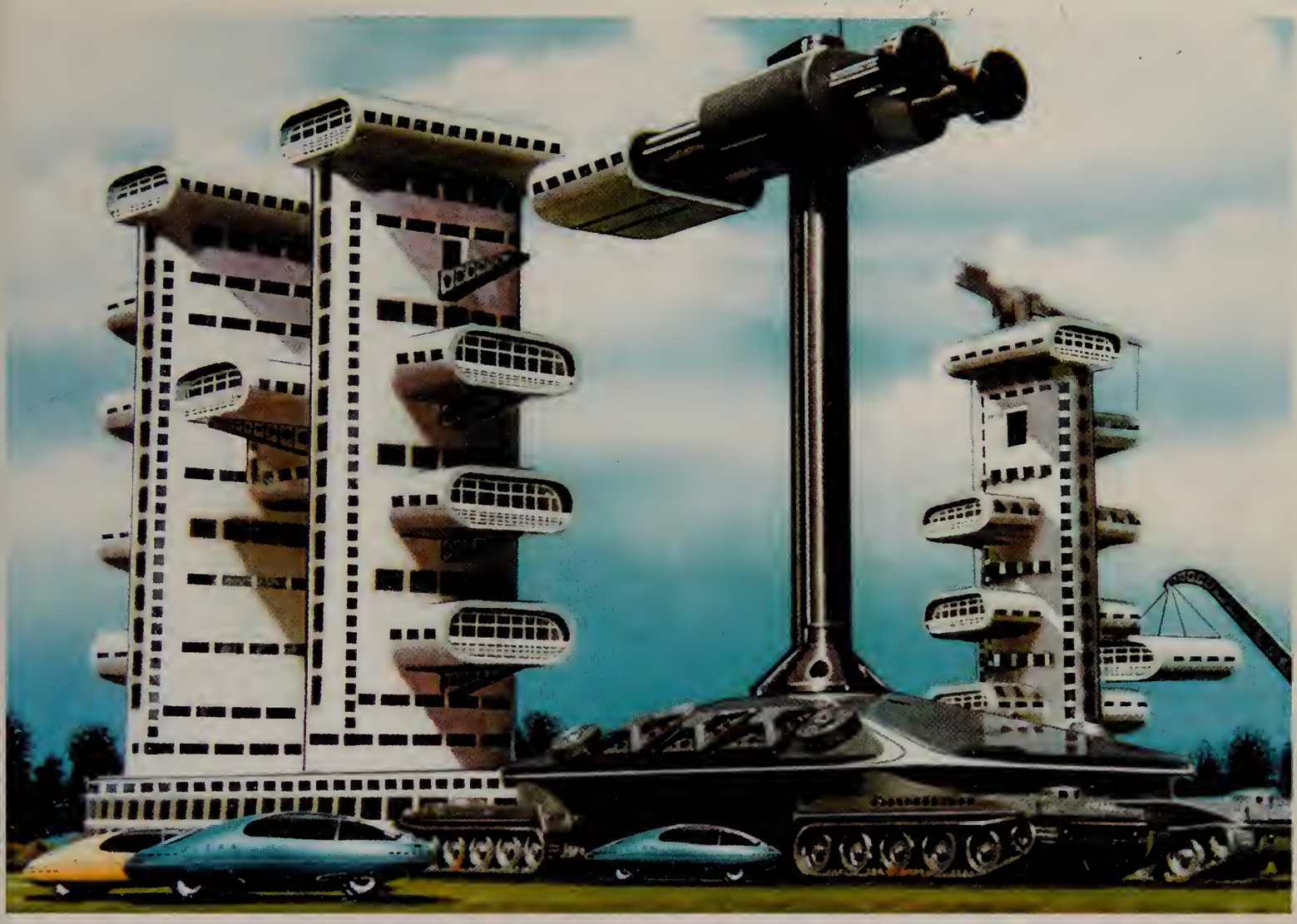
MASS-PRODUCED EXTRUDED DWELLINGS

This demonstrates how lightweight, carbon fiber-reinforced, concrete apartment dwellings could be produced as continuous extrusions, then separated and positioned in place by the mega machine below. The outer shells of these efficient structures serve as photovoltaic generators.



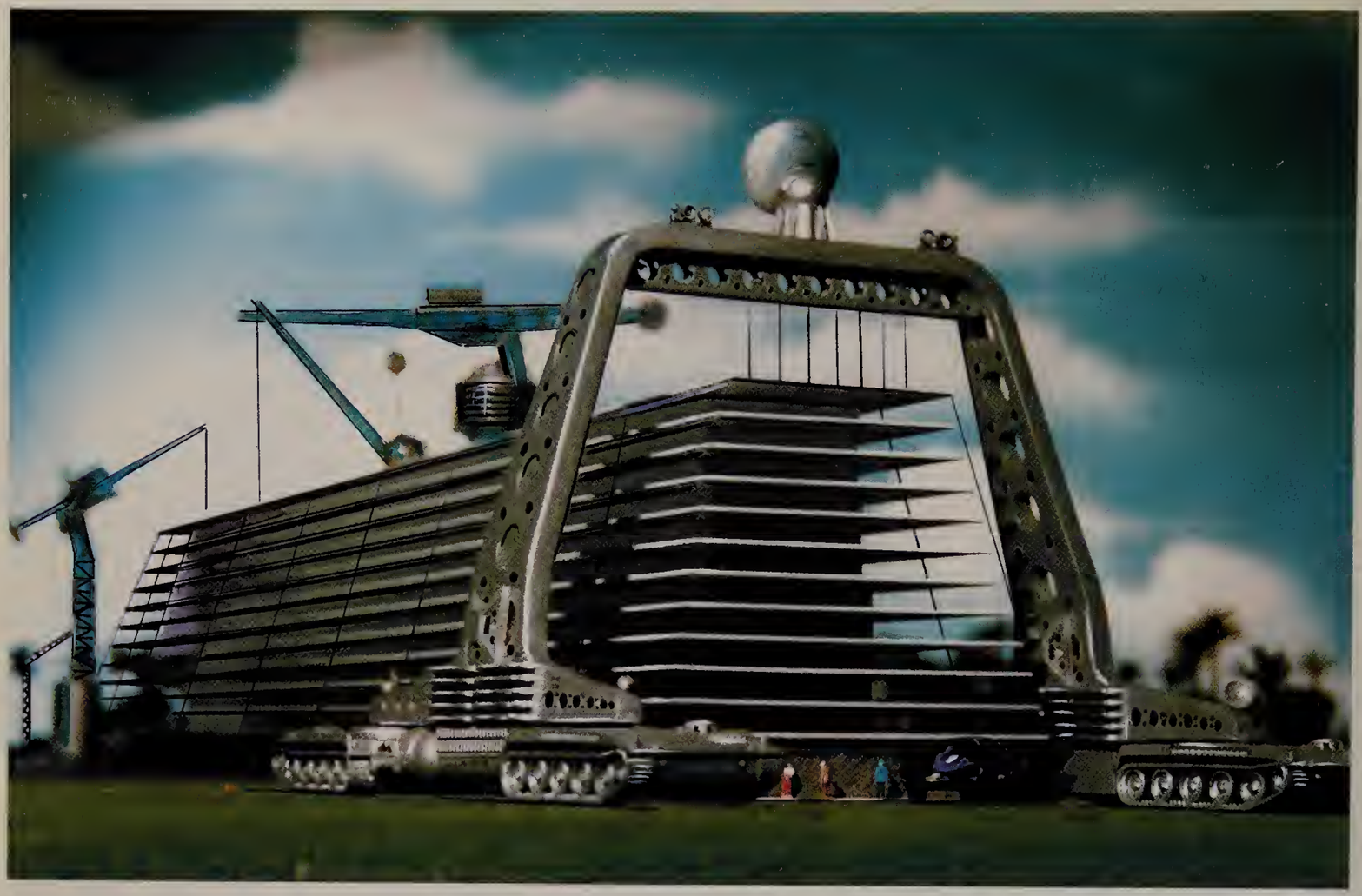
LIFT AND POSITIONING CRANE

This automated machine places these prefabricated dwellings on site where they can be positioned in place by the machine shown on the adjacent page. It is designed to handle structures of many different configurations.



MEGA MACHINES

The machine in the foreground is a multi-function unit. As shown, it lifts and inserts prefabricated housing components into a support structure. Although these "prefab" ufnits will be composed of standard components, they will be of a modular design of such diverse array as to allow maximum individual expression in interior design and décor.



AUTOMATED CONSTRUCTION SYSTEMS

The construction of these industrial and research complexes can be carried out by robotic equipment that receives instructions via satellite. These construction crews of the future consist of automated cranes that travel along the buildings' length, installing floors, windows, curtain-walls, roofing, and other components from the ground up, entirely free from human intervention. These devices will contain self-monitoring sensors to minimize industrial accidents or collisions with other devices or living beings.

CHAPTER 9

WHEN GOVERNMENT BECOMES OBSOLETE

GOVERNMENTS TEND TO ENACT many different kinds of laws in an attempt to control society. However, we find no evidence of a deliberate plan by any government to design a sustainable and workable social system to uplift the lives of everyone rather than those holding high positions in the established order.

Visionaries have sought to improve the lives of people by instigating changes within the established social order. The Semanticists called for improvements and clarifications of meaning in our language. The Communists advocated state-ownership and the end of capitalism and human exploitation. The Fascists created a dictatorship of the rich and powerful. Socialists called for a re-ordering of our priorities to serve humanity by a more equitable distribution of existing resources. Religious groups crusade for a return to simpler times, to family values and the teachings of their charismatic leaders. We call for the establishment of scientific scales of performance applied to the social system for the benefit of all people.

With the application of the methods of science to the social system people would have a better understanding of nature and the symbiotic process of which we are an integral part. This could help to provide an understanding of the interrelationship between ourselves and nature and prevent the over-exploitation of the land and sea.

Many people believe that government leaders bring about change with a deep concern for the well-being of their citizenry. Nothing could be further from the truth. Nor did past shifts in society come about as the results of changes in the schools or the home. All established government systems tend to preserve and uphold their own interests and power-base.

The real forces responsible for change have more to do with unforeseen, external events or biosocial pressures that physically alter our environment and established social arrangements: for example, the infusion of machines and processes that replace people and remove their means of making a living, adverse natural conditions of drought, flood, storm, and earthquake, manmade disasters of economic oscillations, or some outside threat of hostile nations.

The industrial revolution did more than move the centers of population from small farms to large cities. It changed how we relate to our communities. World War II changed the accepted roles of women in this country. Vast droughts and wars in Africa today are

moving whole populations from their ancestral, tribal lands into the cities, destroying entire cultures almost overnight.

Laws are, at best, attempts to placate or control a population. They work only sporadically.

Another effective method designed to control human behavior is the use of early indoctrination towards a given set of values, sometimes manifesting itself as patriotism, propaganda in the national interest, or outright nationalism. In this way, most of the citizenry will tend to support an existing government, while unaware that other options are available.

Yet another safeguard enacted by the politicians is the perpetuation of the concept of responsibility, that we are all in effect responsible for our own shortcomings, failures, and misfortune, when in fact — in accordance with natural laws that govern all activities on Earth — most of our actions are determined by the circumstances that surround us. Many of our so-called free choices have been greatly influenced by the culture and the surrounding values of our times.

All manmade laws are developed to preserve the established order and to protect people from one another in instances such as deceptive business practices, false advertising, theft, and crimes of violence. This calls for constant monitoring of the populace, yet even so the laws are continuously violated. Such problems are largely generated by social insufficiency, not always by the fault of individuals. People cry out for laws for the relief of hunger, poverty, war, oppression and scarcity, but the answer must lie in removing the conditions that are responsible for these problems. There is so much economic deprivation and insecurity — even in the most affluent nations — that no matter the number of laws enacted, the problems persist. Even the legislators who design our laws have permitted gross violations, in some cases breaking the law themselves.

The need for human rights is a result of a scarcity-oriented society. This may be seen with such elements as air and water. Although both are precious necessities to our well-being and survival, there are very few laws regulating how many breaths are taken per hour, because we have such abundance at this time. No-one monitors a gushing spring to see how much water someone takes from it, although fresh water is absolutely necessary for the support of life. If it is abundant, no-one monitors it. In the western US, however, a tangle of laws conflict and overlap on matters to do with agricultural and fishing rights to fresh water.

When a nation creates a set of laws to legislate human behavior, the vast majority of legislators are unaware of the fundamental factors responsible for the need in the first place. All of nature is subservient to natural law. Natural law cannot be violated without serious consequences to the individual, or to societies. Natural laws dominate all living systems. For example, without water, sun, or nutrients, plants and animals would die. In an environment of scarcity, hunger, and poverty, human behavior will adapt itself accordingly.

Where the laws do not correspond to the nature of the physical environment, they will be violated. Look at the myriad of moral codes that attempt to suppress biological sex drives. Eventually, with a deeper understanding of natural law and the effects of social and

cultural influences on human behavior, we may begin addressing the real problems rather than punishing those who transgress.

In a resource-based economy social responsibility would not be instilled by force, intimidation, or promises of heaven or threats of hell. Protection of the natural environment would not be a matter of fines or penalties for polluters. Safeguards against abuse could be designed into the environment. A simple example of this may be seen in the proposed design of the cities of the future where people have free access to resources without debt. This would eliminate the need for theft. Such measures are clearly not a matter of passing and enforcing laws to prevent and punish abuse. Rather, they are a means for designing the flaws out of any social venture, thus eliminating the need for many laws.

Paper proclamations carry very little weight in the real world. All such attempts at social order are BS, Bad Science. Not long ago black Americans did not have access to public water fountains, although there existed so-called constitutional guarantees. Any number of similar examples may be cited of the violation of so-called preferred rights.

A society with human concern "designs out" the need for laws and proclamations, by making all things available to all people, regardless of race, color or creed. When governments make laws, we are led to believe that these laws are made to enhance people's lives. In truth, laws are byproducts of insufficiency.

With an increase in population in excess of available resources, values and behaviors change. With scarce resources, management and allocation become stringent. A set of laws evolve which correspond to these changed conditions. Tracking a culture's evolution reveals the events and environmental influences that determined its values, habits, outlooks, beliefs, and social conduct. For example, if an outbreak of disease reduced the male population by 80%, laws regarding sexual behavior and marriage would undergo vast changes.

We long to be free of flawed, corrupted human thinking and emotions, which have made a graveyard of half of the world — in spite of all the laws, paper proclamations and religious teachings intended to preserve and promote the democratic process in all areas of our monetary-based world economy. Even the United Nations, our most enlightened organization, is motivated to a large extent by self- and national-interests, rather than by the overall good of humankind.

As we begin the transition to a fully cybernated process of governing human affairs, newer technologies will be installed that remove human error from the political bureaucracy. These machines could provide the governing bodies with information rather than opinion, thus considerably reducing the influence of bias and irrational or purely emotional elements in the overall management of human affairs. In an emergent, developing social arrangement not yet established, the rules of human conduct will undergo drastic alterations and changes.

A world-based resource economy could bring about vast changes in human and interpersonal relations, without the enactment of laws. It could introduce a set of values relevant to the needs of all people. A world-based resource economy utilizes all of the world's resources and technical information as the common heritage of all nations, for the benefit of all. This is the unifying imperative. Once installed, the world could witness an end to the

need for armaments, war, drugs, greed, and all of the other problems brought about by the endless pursuit of money and power.

Humans require an education system that teaches process and analytical skills rather than incoherent facts. Dialogue would replace debate. Semantics would be a core skill to greatly improve human communication. Students would intelligently evaluate a situation and access relevant information rather than solve stock problems. It is not that they suddenly become better or more ethical, but that the conditions responsible for hostile and egocentric behavior are no longer present.

Today we try to control human behavior by enacting laws or signing treaties without changing the physical conditions responsible for aberrant behavior. When the earth's resources become the common heritage of all people the necessity for these irrelevant laws and social contracts vanish.

As for "who" will govern, the more appropriate question is "how will people be governed?" People do not have to be governed or require leaders unless they are ignorant, captive, wage slaves, or agree to be subjected to a dictator. If the free enterprise system does not include job security, medical care, and all the other necessities that secure the position of the population as a whole, a wide range of conflicts and unmanageable human behaviors results, no matter how many laws are passed.

No "one" will decide who gets what. Perhaps the closest analogy within our present culture would be the public library, in which anyone has access to any book of his or her choice. Goods and services could be made available in a similar manner across the entire spectrum of the social system. Unfortunately, we are in the habit of thinking that someone has to make the decisions regarding our needs. This would not be the case in the operation of a cybernated, resource-based society.

In the near future, due to such advances in technology as artificial intelligence, cybernation, and nanotechnology, we could achieve a global community and a common vision for humanity. Computerized technology will have greater success uniting people and eliminating scarcity than have all of the world's religions and democratic ideals combined. We can now transcend the limitations of a monetary system, and in time may outgrow our need for politicians and artificial, manmade laws intended to preserve and perpetuate the status quo. *AI* could regulate production, transportation, and all burdensome and monotonous tasks — ***but not people.*** A highly integrated complex of computers that serves, ***but never enslaves,*** humankind could carry out the major tasks of decision-making and environmental management.

I must again emphasize that this approach to global governance has nothing whatever in common with the present aims of an elite to form a world government with themselves at the helm, and the vast majority of the world's population subservient to them. This newer vision of globalization empowers each and every person on the planet to be all that they can be, not to live in abject subjugation to a corporate governing body.

The question is, can we grow beyond thinking that "someone" has to make our decisions for us?

CHAPTER 10

WHO WILL MAKE THE DECISIONS

THROUGHOUT HISTORY, the societal decision-making process has gone though a number of changes. At one time, primitive tribes and their ruling chieftains and kings decided upon a set of laws, beliefs, and mores designed to support and defend the ruling oligarchies. As primitive cultures joined together, possibly for mutual protection, the chieftains of the various tribes shared some decision-making.

With the advent of nations, councils were appointed to participate in decision-making, to prevent any one of the leaders from dominating. However, the less privileged were not included in this process. As the ruling classes imposed greater hardships on their subjects through taxation and other abuses of power, uprisings by the oppressed people, intrigue, sabotage, and assassinations forced changes in the laws of the land. Governing bodies were then appointed to carry out and uphold laws.

Although wealth has always "bought" political offices, it was not until the beginning of the nineteenth century that financial interests began in earnest to play a leading role in inappropriate decision-making. Many politicians will use every legal means of deception known to consolidate their position, in the process repeating old slogans that have been used for centuries — "a return to family values," "to serve God and country," and other verbal expressions of undefined goals. They talk around every significant issue without saying anything of content, placing a great deal of emphasis on the role of law and order in government and on international agreements. They enact new laws to try to control the behavior of people and if these don't work, they resort to a tried-and-true method — the use of force, of boycotts and blockades. But none of the methods used seems ever to address the root cause.

Most people still believe that all that is needed to set things right is to replace the incompetent and corrupt officials in government with decent men and women of high moral character. Although we occasionally find politicians of sincere intent, they too are usually unable to find answers to the problems that exist at any given point in time. Human systems fail, obviously, to serve the needs of humanity. This is true across the entire span of human administration: the church, the government, the military, and the banks. In times past, most of the social designs were unsuccessful for the majority of people because their designers were unable to transcend the limits of their own environmental conditioning. We tend to bring our past into the present and project it into the future.

Today, the laws that govern our society are not based on any truly comprehensive studies. They are based on opinions and traditional practices. For example, our current approach to dealing with an increase in crime is to build more prisons rather than to alter the conditions responsible for socially offensive behavior. In a recent discussion with criminologists, it was pointed out that if our crime rate continues at its current level, more than half the US population will be in prison by the year 2010. The other half may well have to guard them. Rather than depend on a failed system of punishment or incarceration after the damage has been done, a more effective approach to solving our problems would be to shift our attention to the scourges of poverty, malnutrition, poor role models, violence in the media, and stresses in family life, while making more of an effort to teach people how to resolve conflict without the use of physical force.

The discovery of scientific principles enables us to validate and test many proposals. If someone claims that a particular structural element can support a specific number of pounds per square inch, this hypothesis can be tested and either substantiated or negated based upon the test results. It is precisely this process of testing which enables us to design and construct bridges, buildings, ships, aircraft, and all other mechanical wonders.

In the new social design as outlined in this book, scientific and analytical principles can be applied not only to industry and construction, but also to all of the personal components of society. This may eventually lead to the application of more scientific resources to the study of human behavior. The most difficult aspect of the redesign of a culture is the seemingly undemocratic approach to the problem. By what authority does any group presume to impose an arrangement of social affairs on others who may be living in the arrangement?

This brings up three questions of primary importance to the redesign of a culture:

1. For whom is the culture designed?
2. What ends is it designed to serve?
3. Who will benefit, the rulers or the ruled?

Thought history, social affairs have either been pre-arranged, or have eventually worked out to benefit the power elite and money interests. Even in so-called democracies this has been the case. People fear a planned social system may not serve their interests. They perceive a very real danger that the introduction of any new social arrangement carries with it the possibility of the development of a new elite.

If a particular religious group were to design a society, it would quite naturally reflect the group's belief's — the "will of the people". The majority of this group would democratically agree that theirs is a good social design. The atheist, agnostic, Hindu, Muslim, and all others not represented would naturally object. What is really needed is a system to determine the most appropriate direction that will be agreeable to all. As difficult as this may appear, it can be done.

Today we have a decentralized system of decision-making, and the decision-makers are seldom aware of problems in regions outside their immediate vicinity. Those of us in

sub-tropical Florida have difficulty understanding water rights in Arizona. A Berger of Morocco would be challenged to design a health plan that matched the life styles of people in Norway. Each of us must participate. And we need verifiable and recent information on which to plan.

When computers eventually have their electrical sensors extended into all areas of the social complex, we will be able to return to the centralization of decision-making. In a global resource-based economy the decisions would not be based on local politics but on a holistic problem solving approach. The earth itself, and all the life on it, must be seen as constituting a single system.

This centralized system could be connected to research labs and universities, so that all data is monitored and updated constantly. Most of the technology to allow such infrastructure management is currently available.

For example, when the electrical sensors are extended into the agricultural region, these computerized systems would be able to manage and control all of the agricultural requirements by monitoring the water table, insects, pests, plant diseases, soil nutrients, and so forth.

Computers and artificial intelligence will serve as the catalyst for change. They will be able to establish scientific scales of performance. It is doubtful that in the latter part of the twenty-first century people will play any significant role in decision-making. Eventually the installation of *AI* and machine decision-making will manage all resources serving the common good.

Computers as decision-makers will always scan for new information and methods of conserving resources to coincide with the carrying capacity of each geographical region. This will result in a more humane and meaningful approach for shaping tomorrow's civilization, one that is not based on the opinions or desires of a particular sect or individual.

In the event of a breakdown or regional or national emergency, the special information and already-developed alternative plans for any of the known types of catastrophes that occur would be available, just as military contingency planning is done today.

All decisions would be made on the basis of a comprehensive resource survey and the availability of energy or existing technology — not on the advantage to be gained by any nation or select group of people. This resource survey would coincide with the carrying capacity of each geographical region of the global environment